

Information Visualization and Visual Analytics (M1522.000500)

# Perception and Visual Patterns

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Seoul National University

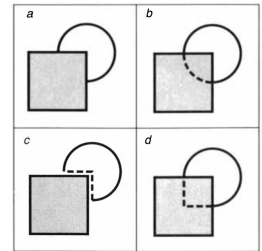
## Visual Patterns

### Visual Patterns

- Key to information visualization
- Understanding patterns to build knowledge
- Using vision to think (finding patterns)
  - Patterns showing groups
  - Patterns showing structure
  - When are patterns similar?
  - How should we organize information on the screen?

## Gestalt Laws

- Principles of perceptual organization
  - the **whole** is different from the **sum of its parts** – central tenet of Gestalt psychology
  - how smaller objects are grouped to form larger ones
  - "gestalt": German for "pattern/form/shape" (or "configuration")
  - "leaving us with a set of *descriptive* principles, but without a model of perceptual processing"
  - rules themselves still very useful
- The Principle of Prägnanz - fundamental principle of gestalt perception
  - law of simplicity, law of **good** figure
  - tend to order our experience in a manner that is **regular, orderly, symmetric, and simple**
  - simplest possibility wins



OBSURED FIGURE illustrates the idea of *Prägnanz*. Given an ambiguous pattern (a), observers perceive simple shapes (b) instead of complex ones (c, d).

image source: URL: <https://www.jstor.org/stable/10.2307/24997014>

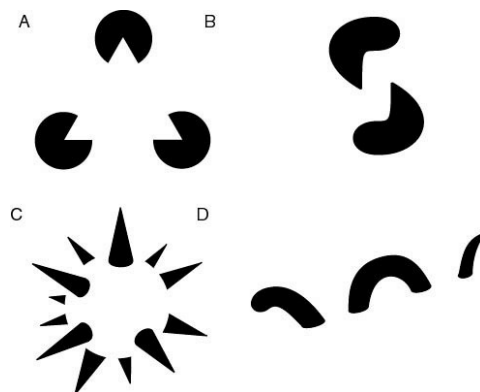
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## Principles



*Emergence*

the dog is perceived as a **whole**

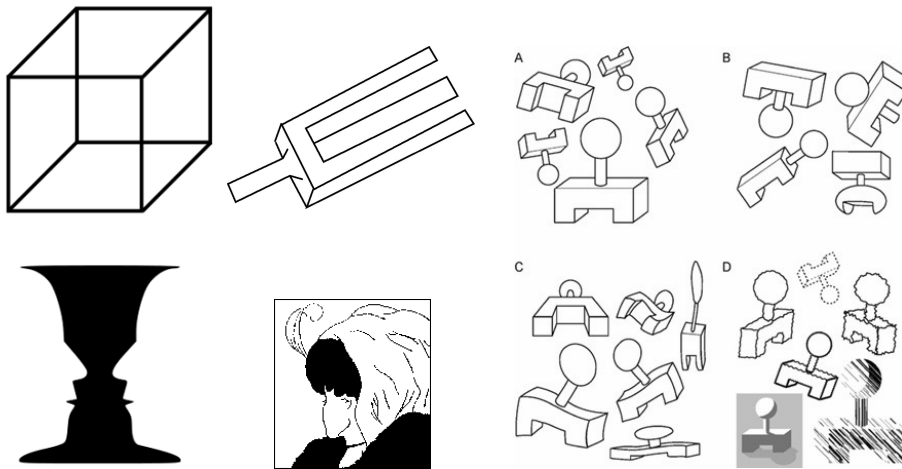


*Reification*

the experienced percept contains **more explicit spatial information** than the sensory stimulus on which it is based

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## Principles



Multistability Perception

Invariance

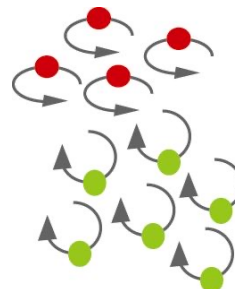
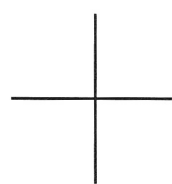
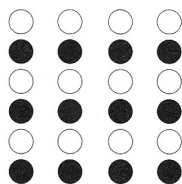
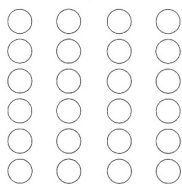
<http://www-personal.umich.edu/~esrabkin/pics/ThreeProngedTuningFork.jpg>

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## Gestalt Principles

## • Grouping

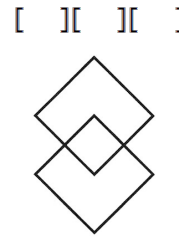
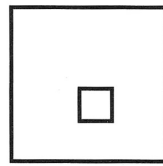
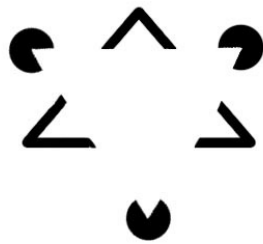
- **Proximity:** tendency of elements to be associated with nearby elements
- **Similarity:** tendency of elements to be associated with similar elements
- **Continuity:** preference for continuous, unbroken, smoothest contours with the *simplest possible physical explanation*
- **Common Fate:** things moving together



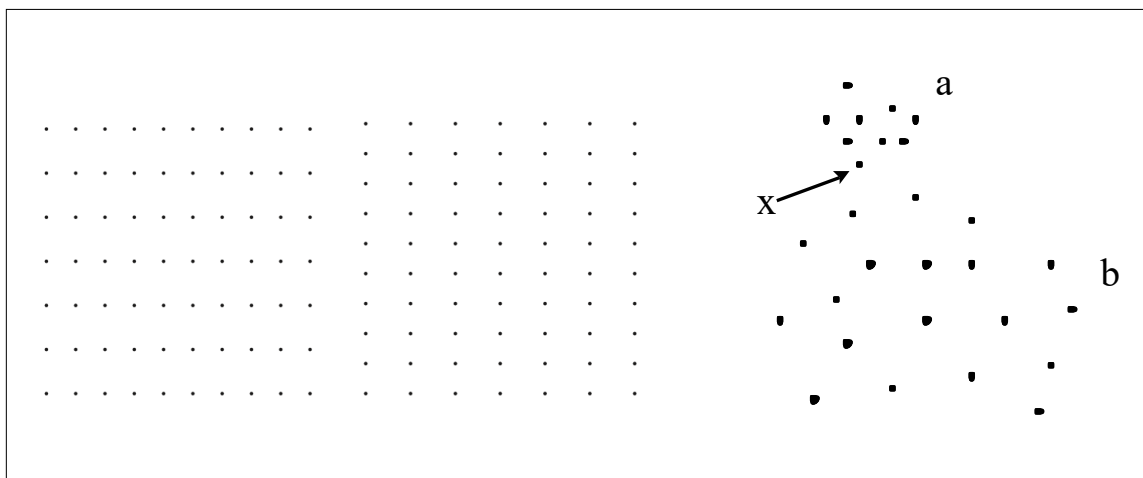
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## Gestalt Principles

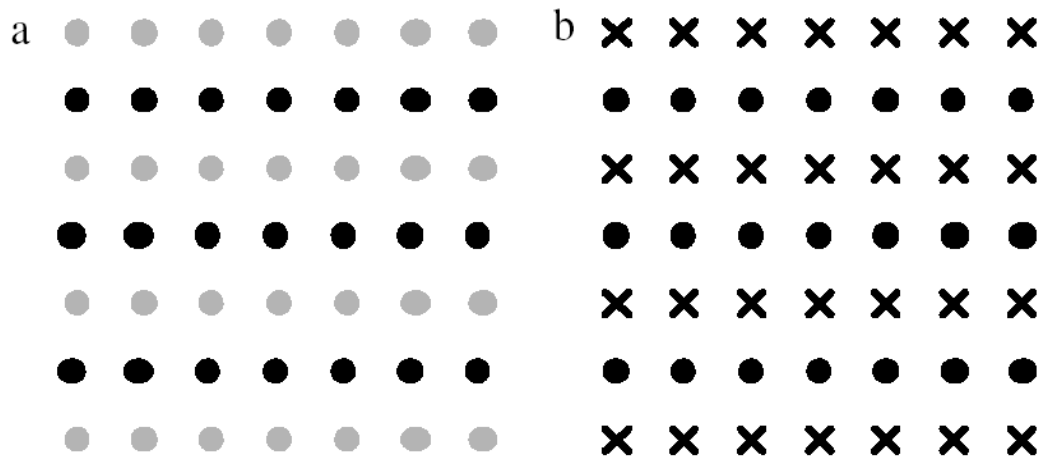
- Perception of Forms
  - **Closure**
    - form complete, closed figures to increase regularity
  - **Figure and Ground**
    - smaller one as figure, larger one as ground
  - **Symmetry**
    - when we perceive objects, we tend to perceive them as *symmetrical shapes that form around their center*.
    - symmetrical images are perceived collectively, even in spite of distance



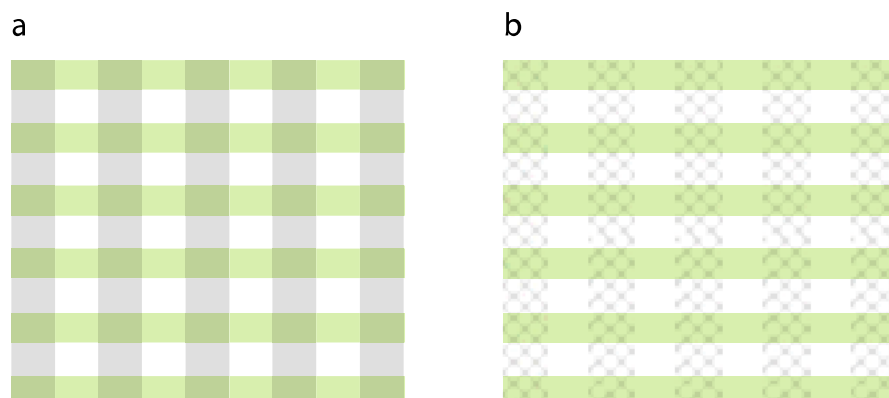
## Proximity (The principle of spatial concentration)



## Similarity



## Similarity

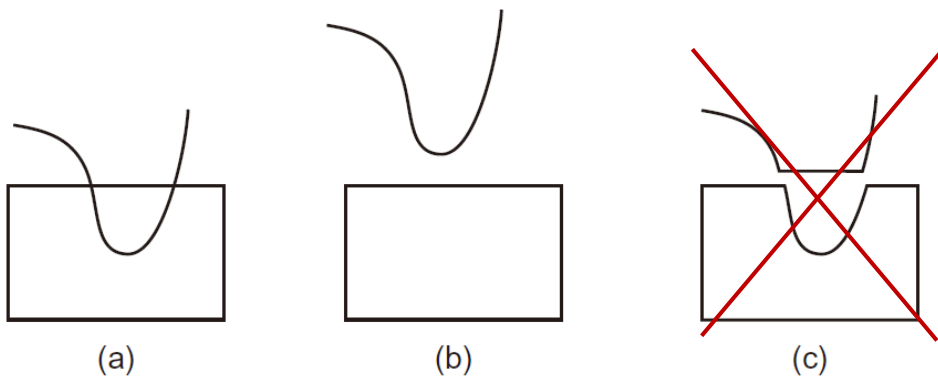


Proximity and Similarity



Continuity

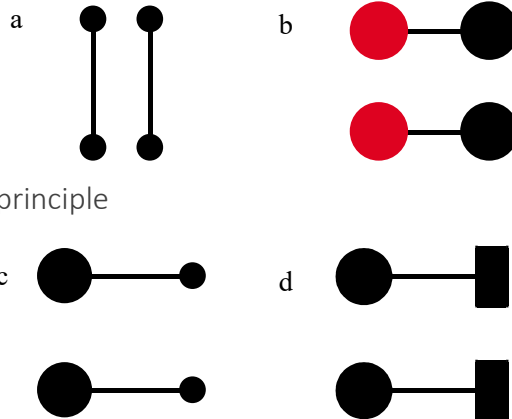
- Visual entities tend to be smooth and continuous



## Connectedness

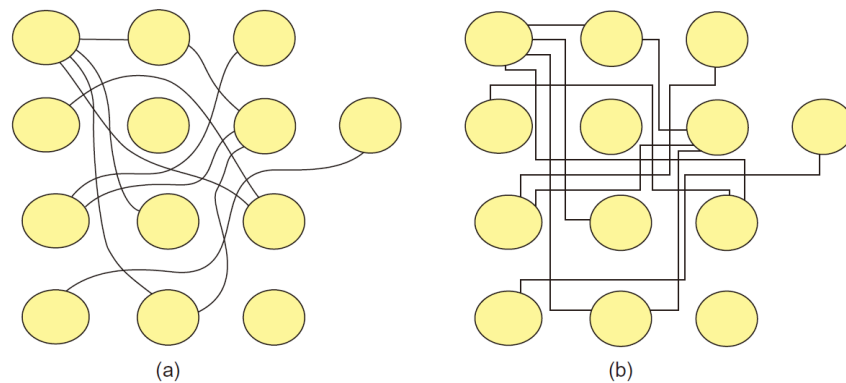
- assumed in Continuity
- can overrule size, shape

- Connectedness is a powerful grouping principle that is stronger than (a) proximity, (b) color, (c) size, or (d) shape.



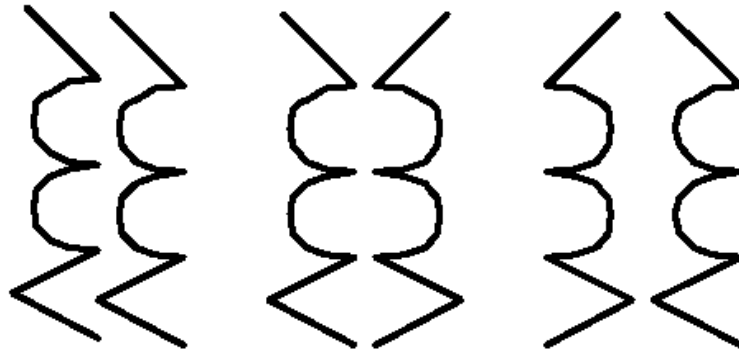
## Continuity in Diagrams

- Connections using smooth lines
  - easier to perceive than using lines with abrupt changes



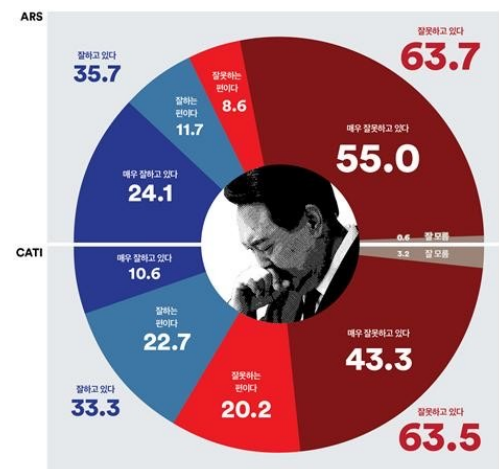
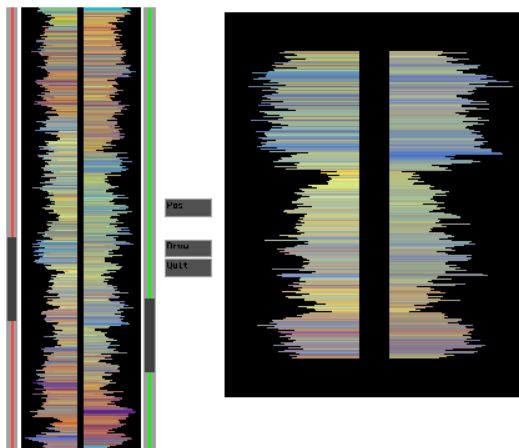
## Symmetry

- sense of a **holistic** figure
- emphasizes relationships



## Symmetry

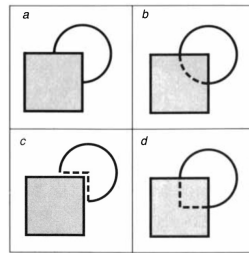
- symmetry to show Similarities between time series data





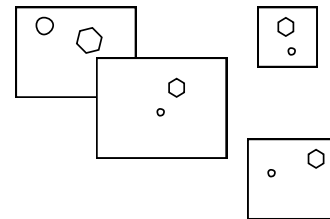
## Closure

- Prefer closed contours
- Overrule proximity



OBSCURED FIGURE illustrates the idea of *Prägnanz*. Given an ambiguous pattern (a), observers perceive simple shapes (b) instead of complex ones (c, d).

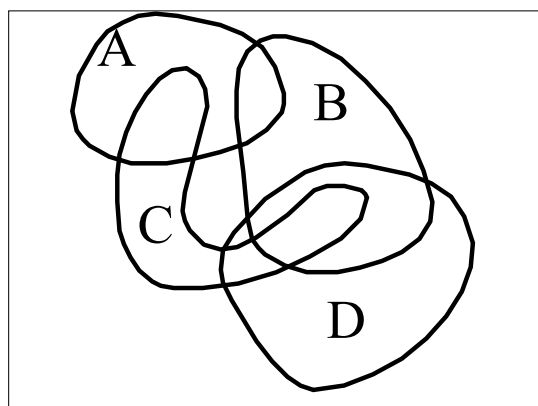
[ ][ ][ ][ ]



- Segment visual field
- Provide reference frames

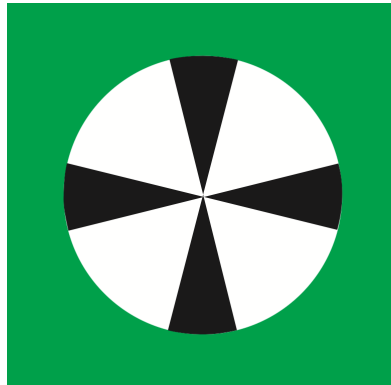
## Closure

- Closed contours to show set relationship



## Figure-Ground Effects

- smaller components perceived as objects



The black areas are smaller and therefore more likely to be perceived as an object.

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## Figure-Ground Organization

- Figure-Ground Organization is fundamental to perception
  - Either side of the pattern on the left can be perceived as figure or as ground
  - Although the two shapes on the right share the same contour, they seem very different



image source: URL: <https://www.jstor.org/stable/10.2307/24997014>

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## Gestalt Laws

### Laws of Grouping

#### NO GROUPING



#### GESTALT LAWS OF GROUPING



PROXIMITY



SIMILARITY



CLOSURE



GOOD CONTINUATION

#### NEWLY PROPOSED LAWS



COMMON REGION



CONNECTEDNESS

image source: URL: <https://www.jstor.org/stable/10.2307/24997014>

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## Gestalt Laws

### Natural Camouflage

- It shows how the laws of grouping, such as similarity, proximity and connectedness, can conceal animals.
  - The ponies cannot easily be distinguished from the background.

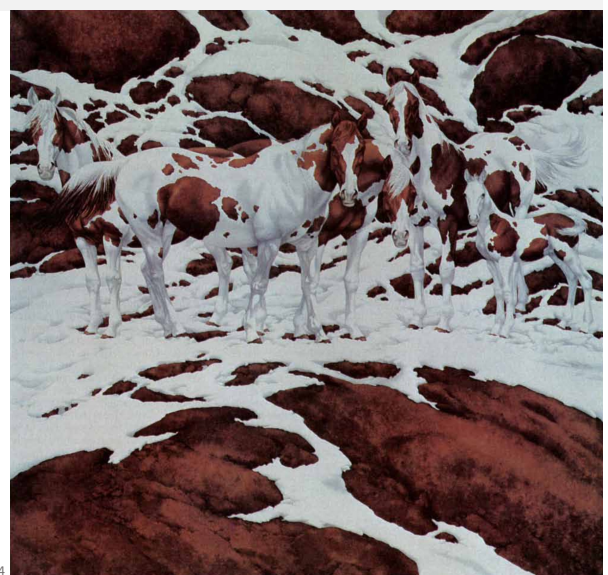


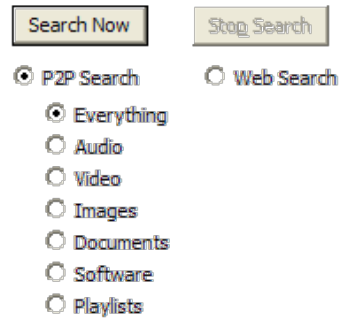
image source: URL: <https://www.jstor.org/stable/10.2307/24997014>

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## Proximity Examples



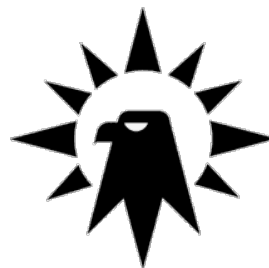
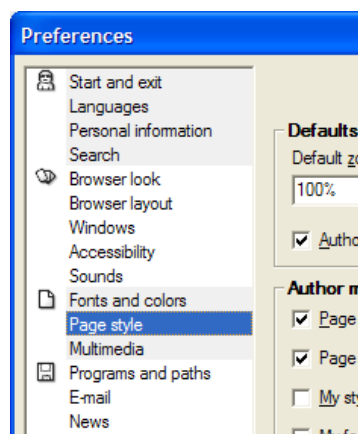
MTV Music Awards 2002



[http://www.interaction-design.org/encyclopedia/gestalt\\_principles\\_of\\_form\\_perception.html](http://www.interaction-design.org/encyclopedia/gestalt_principles_of_form_perception.html)

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## Similarity Examples



[http://www.interaction-design.org/encyclopedia/gestalt\\_principles\\_of\\_form\\_perception.html](http://www.interaction-design.org/encyclopedia/gestalt_principles_of_form_perception.html)

<http://graphicdesign.spokanefalls.edu/tutorials/process/gestaltprinciples/gestaltprinc.htm>

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## Grouping

- Help users parse the display into sub-units
  - Rely on Gestalt principles
  - Avoid explicit grouping

A dialog box titled "Composite" with four distinct sections, each enclosed in a rounded rectangle. The sections are: "Source 1" (Document: concept.1, Channel: RGB), "Mask" (Document: concept.1, Channel: RGB), "Source 2" (Document: concept.1, Channel: RGB), and "Destination" (Document: New, Channel: New). Each section has its own "OK" and "Cancel" buttons.

A dialog box titled "Composite" with a flat layout. It has two columns of labels: "Document" and "Channel". The rows are: "Source 1" (concept.1, RGB), "Mask" (concept.1, RGB), "Source 2" (concept.1, RGB), and "Destination" (New, RGB). The "OK" and "Cancel" buttons are on the right.

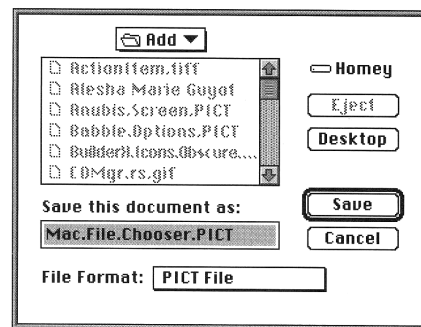
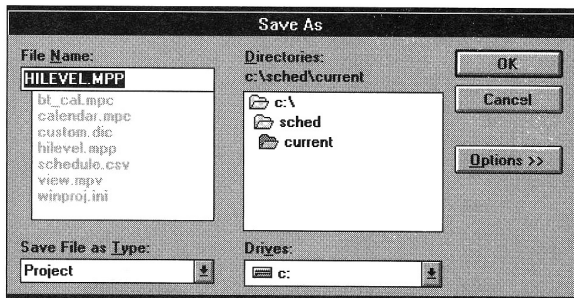
## Hierarchy

- Provide a context for each piece of information
  - Example: distinctive style for labels

A dialog box titled "Pizza: Pizza Pie Properties" with a hierarchical structure. It is divided into "BOTTOM STUFF" and "TOPPING STUFF". "BOTTOM STUFF" includes "Style" (Thin, Thick) and "Sauce" (Regular, Spicy, Pesto). "TOPPING STUFF" includes "Toppings" (Meat: Ham, Bacon, Sausage; Veggies: Olive, Pepper, Onion; Other: Cheese, Pesto, Anchovy) and "Options" (Red Peppers, Extra Cheese). The "Apply" and "Reset" buttons are at the bottom.

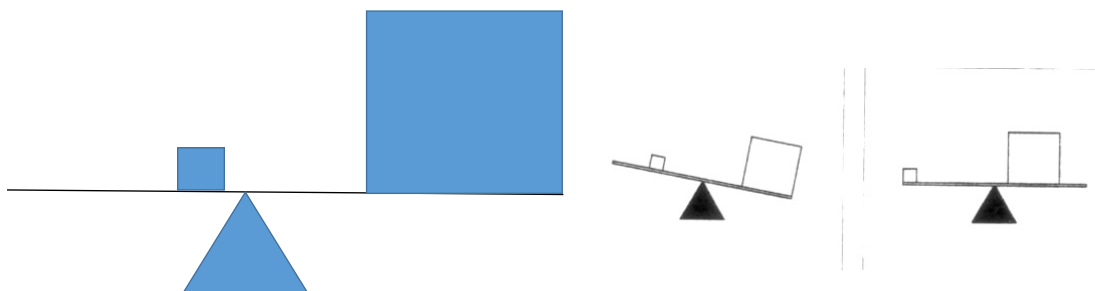
## Relationship

- Parallel between presentation and relationship
  - Example: present folder before its content



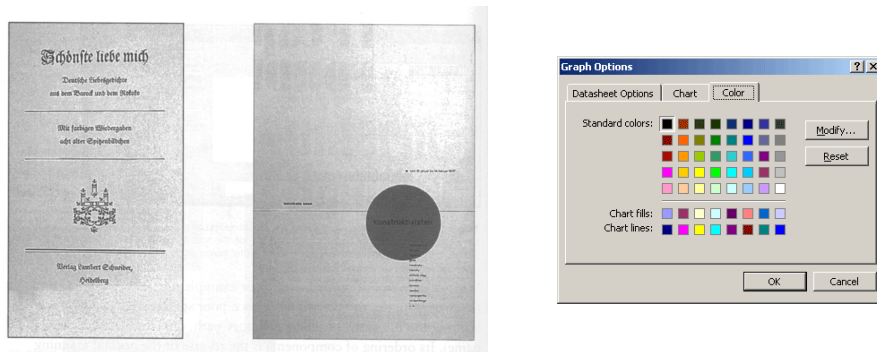
## Balance

- Harmonious global arrangement
  - Can be symmetrical or asymmetrical
  - smaller elements can offset the **visual weight** of larger elements, how?



## Balance

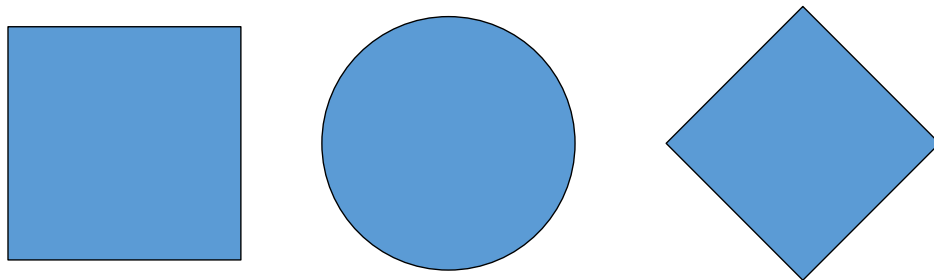
- Harmonious global arrangement
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## Human Size Perception

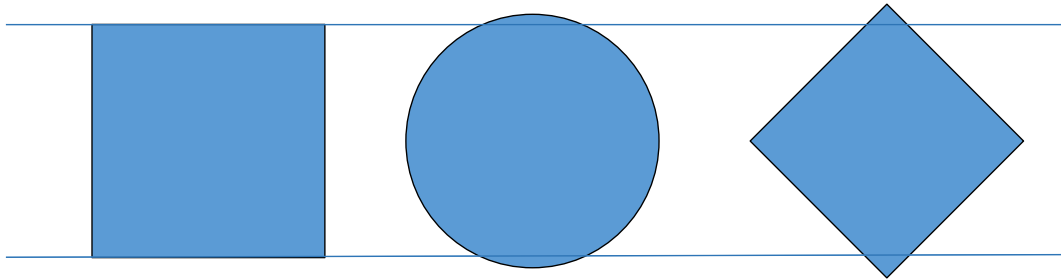
## Human Size Perception

- Straight edges appear larger than curved edges
- Curved edges appear larger than sharp edges
- Optical adjustment



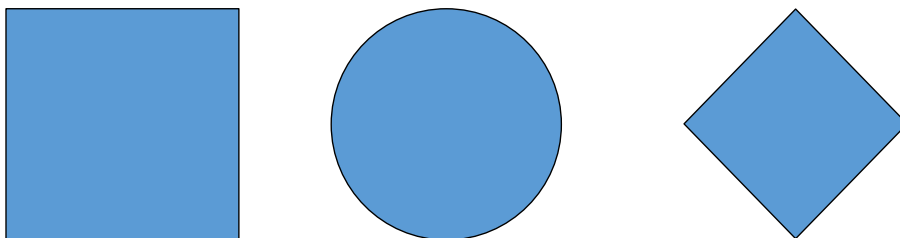
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## Human Size Perception

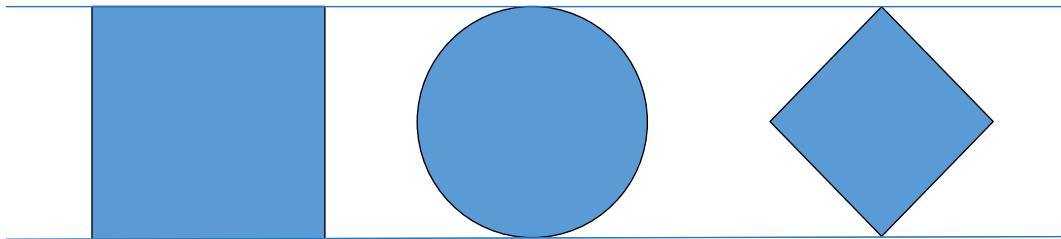
- Straight edges appear larger than curved edges
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### Human Size Perception

- Straight edges appear larger than curved edges
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### How human visual system works

- “What **visual properties** draw our eyes, and therefore our **focus of attention** to a particular object in a scene?”
- **Visual attention**: mechanisms that help determine **which regions of an image are selected for more detailed analysis**
- **Detailed vision** for shape and color is only possible within a small portion of the visual field (1 degree of visual angle, **foveal vision**)
- **fixation-saccade cycle**

## Fixation-Saccade Cycle



[http://www.outofmygord.com/images/outofmygord\\_com/eyetrackingsaccades.gif](http://www.outofmygord.com/images/outofmygord_com/eyetrackingsaccades.gif)

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## Fixation-Saccade Cycle

- **fixation**: brief stationary period when detail information is acquired
- **saccade**: flicking rapidly to a new location during a brief period of blindness
  - saccade takes at least 200 ms to initiate
- repeats 3~4 times each second
- makes seeing highly dynamic
- bottom-up: information from fixation → mental experience
- top-down: current mental states (tasks and goals) → guiding saccades

### Visual Expectation and Memory

- “What do we **remember** about an object or a scene when we stop attending to it and look at something else?”
- role of memory and expectation in seeing
- current state of mind plays a critical role
  - determining what is being seen, what is not being seen, what will be seen next
- postattentive amnesia, memory-guided search, change blindness, inattention blindness, attentional blink

Christopher G. Healey, James T. Enns: Attention and Visual Memory in Visualization and Computer Graphics. IEEE Trans. Vis. Comput. Graph. 18(7): 1170-1188 (2012)

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### Postattentive Amnesia

- Does previewing of stimuli make search faster?
  - Extract detail rapidly on demand?
- Human vision is not an optical camera
  - Region of the most recent focus of attention
  - Detail is only available there

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Can studying a display offers assistance in searching for specific data values?

- traditional search vs. postattentive search

**GREEN  
VERTICAL**

Traditional Search



Wolfe, Klempe, Dahlen, "Post Attentive Vision," JEP: HPP 26 (2), 2000.

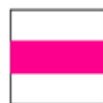
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**WHITE  
TILTED**



Wolfe, Klempe, Dahlen, "Post Attentive Vision," JEP: HPP 26 (2), 2000.  
Christopher G. Healey, James T. Enns: Attention and Visual Memory in Visualization and Computer Graphics. IEEE Trans. Vis. Comput. Graph. 18(7): 1170-1188 (2012)

## Postattentive Amnesia

- Postattentive search was as slow (or slower) than the traditional search
  - with approximately 25-40 *ms* per object
  - studying a display offers no assistance in searching for specific data values
- Implications for visualization design
  - In most cases, visualization displays are novel
  - their contents cannot be committed to LTM
  - **preattentive methods are critical for efficient data exploration**
  - draw attention to areas of potential interest

Christopher G. Healey, James T. Enns: Attention and Visual Memory in Visualization and Computer Graphics. IEEE Trans. Vis. Comput. Graph. 18(7): 1170-1188 (2012)

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## Preattentive Tasks

- visual features that are detected very rapidly by low-level, fast-acting visual processes
- seems to precede focused attention
  - occurring within a single fixation
  - attention plays a critical role in what we see in this early stage
- preattentive tasks: performed on large multi-element displays in less than 200-250 *ms*
  - saccade takes at least 200 *ms* to initiate
- “pop out” of a display
  - easily detected *regardless of the number of distractors*
  - vs. time-consuming visual search
- Target detection, Boundary detection, Region tracking, Counting and estimation

Christopher G. Healey, James T. Enns: Attention and Visual Memory in Visualization and Computer Graphics. IEEE Trans. Vis. Comput. Graph. 18(7): 1170-1188 (2012)

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How many sevens?

2398419309213985874506209348952034809502  
3984210293841909238740129384610329849238  
4265293845013945594858601239480234958728  
4596394058640598239485802394895029348658  
4561024596234851604569828309458673049561  
3045916459086130495298646658956405196809  
5866304598683049561835601830459680345907  
6283486510465183560241620945613045618304  
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Slide Idea from Colin Ware

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Color Makes Them Pop Out

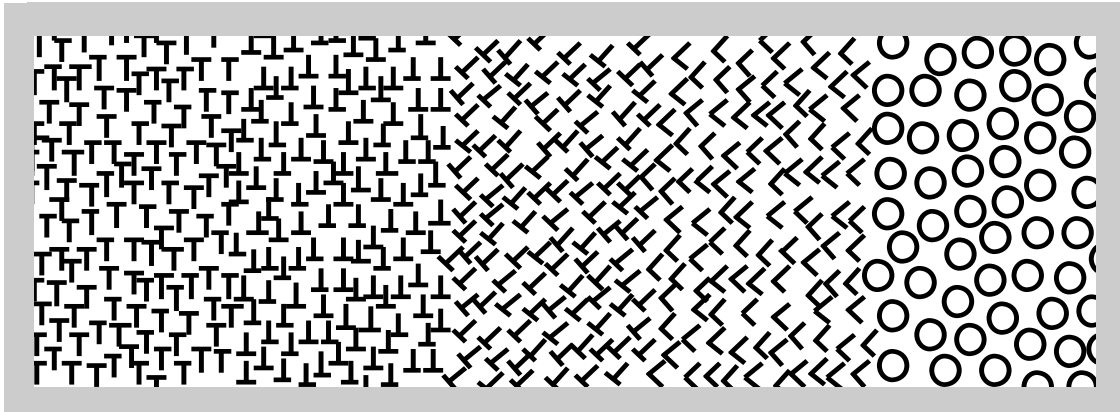
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6283486510465183560241620945613045618304  
5968230459630459860395620349568204385362

Slide Idea from Colin Ware

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## Segmentation



Slide idea from Colin Ware

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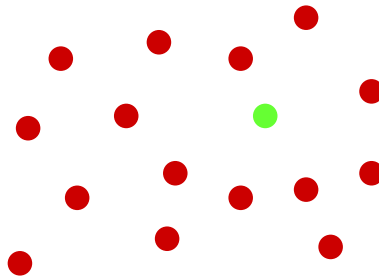
## What Kinds of Tasks?

- Target detection
  - Is something there?
- Boundary detection
  - Can the elements be grouped?
- Region tracking
  - Can a distinctive moving group be traced?
- Counting and estimation
  - How many elements of a certain type are present?
  - Estimate the number of elements with a unique visual feature

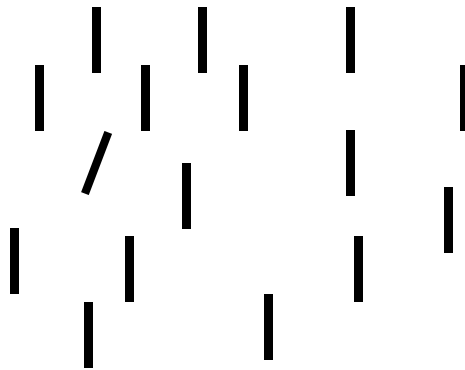
John Stasko

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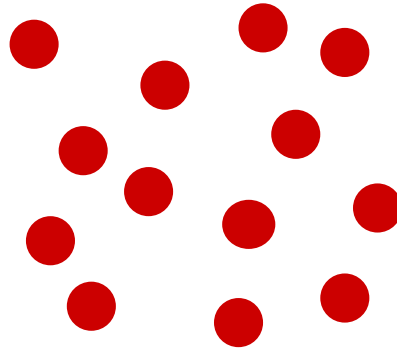
Color-Hue



## Orientation



## Motion

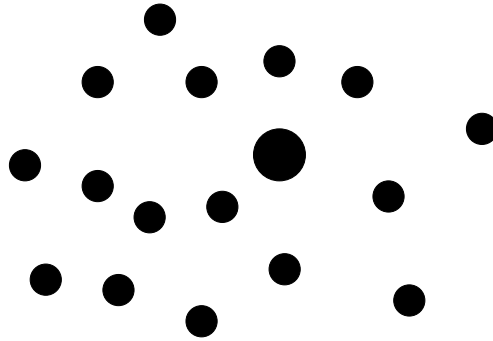


Colin Ware

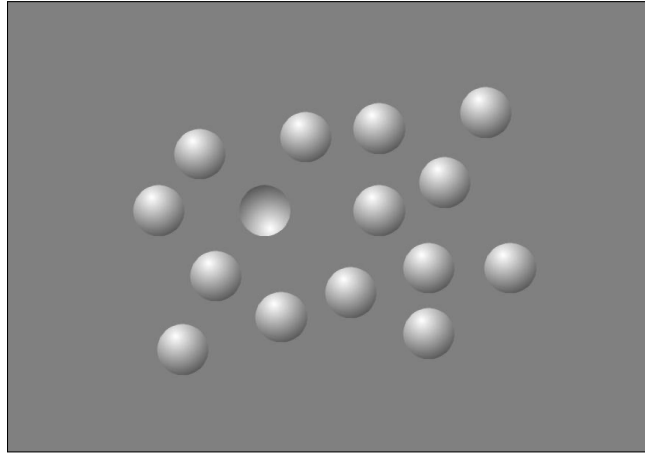
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Size



Simple shading

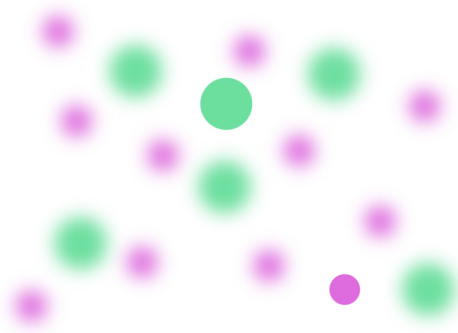


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## Semantic Depth of Field

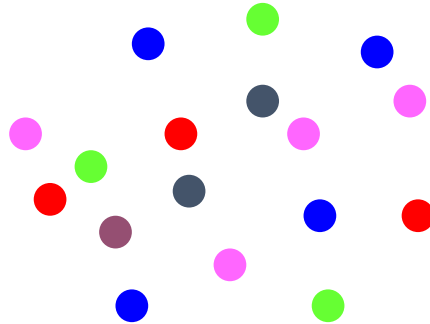


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Surrounded colors do not pop out



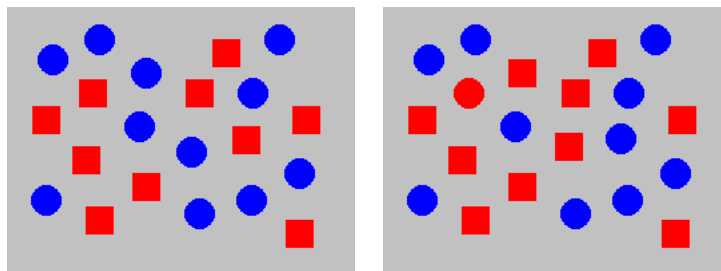


## Laws of preattentive display

- Must stand out on some simple dimension
  - color,
  - simple shape = orientation, size
  - motion,
  - depth
- Lessons for highlighting – one of each

## Hue and Shape

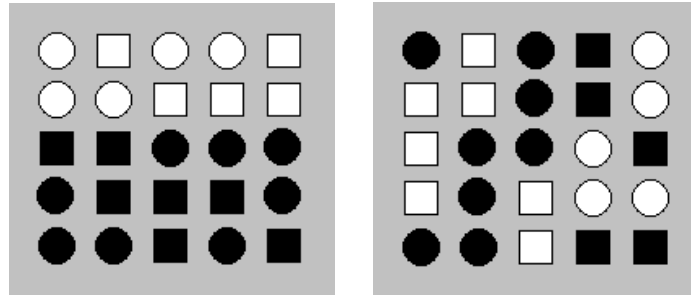
- Determine if a red circle is present



- Cannot be done preattentively
- Must perform a sequential search
- Conjunction of features (shape and hue) causes it

## Brightness and Shape

- Is there a boundary?



- Left can be done preattentively since each group contains one unique feature
- Right cannot since the two features are mixed (fill and shape)

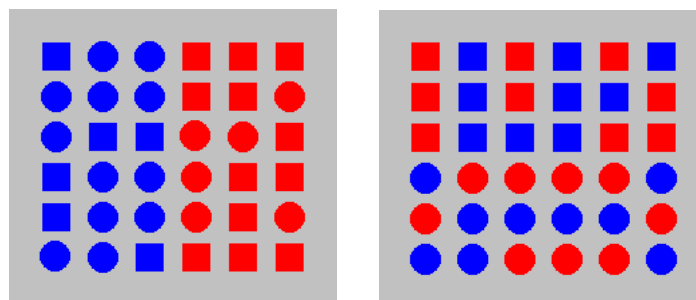
<http://www.csc.ncsu.edu/faculty/healey/PP/PP.html>

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## Hue versus Shape

- Is there a boundary?



- Left: Boundary detected preattentively based on hue regardless of shape
- Right: Cannot do mixed color shapes preattentively

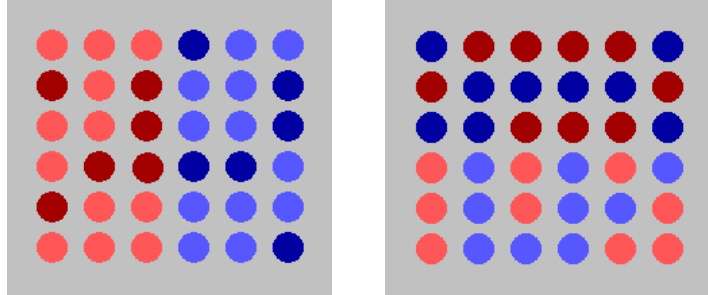
<http://www.csc.ncsu.edu/faculty/healey/PP/PP.html>

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## Hue versus brightness

- Is there a boundary?



- Left: Varying brightness seems to interfere
- Right: Boundary based on brightness can be done preattentively

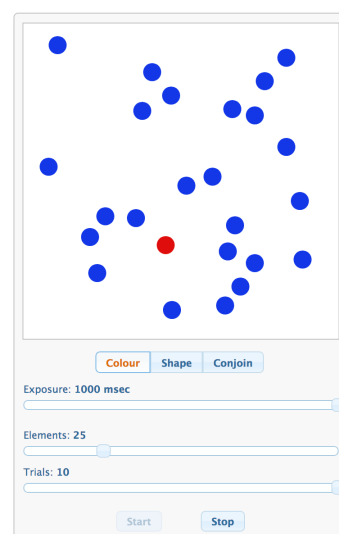
<http://www.csc.ncsu.edu/faculty/healey/PP/PP.html>

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## Preattentive Visual Features

- Perception in Visualization (by Chris Healey, NC State)
- Preattentive Visual Features
  - line (blob) orientation
  - length, width
  - closure
  - size
  - curvature
  - density, contrast
  - number, estimation
  - colour (hue)
  - intensity, binocular lustre
  - intersection
  - terminators
  - 3D depth cues, stereoscopic depth
  - flicker
  - direction of motion
  - velocity of motion
  - lighting direction
  - 3D orientation
  - artistic properties



<http://www.csc.ncsu.edu/faculty/healey/PP/PP.html>

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## Readings

- Perception in Visualization, Christopher G. Healey,  
<http://www.csc.ncsu.edu/faculty/healey/PP/index.html>
- **Attention and Visual Memory in Visualization and Computer Graphics**, Christopher G. Healey & James. T. Enns,  
TVCG, Vol 18(7), pp. 1170-1188, 2012
- Chapter 4, Organization and visual structure, Designing Visual Interfaces: Communication Oriented Techniques, Kevin Mullet and Darrell Sano, Prentice Hall PTR, 1994, ISBN 978-0133033892
- Chapter 5, Visual Attention and Information that Pops Out, Chapter 6, Static and Moving Patterns, Information Visualization: Perception for Design, 2nd ed. Colin Ware, Morgan Kaufmann, 2004, ISBN 1-55860-819-2.

## Credits

- John Stasko, Georgia Tech
  - [www.cc.gatech.edu/~stasko/7450/](http://www.cc.gatech.edu/~stasko/7450/)
- Colin Ware, U of NH
  - <http://ccom.unh.edu/vislab/VisCourse/index.html>
- Tamara Munzner, UBC
  - [people.cs.ubc.ca/~tmm/courses/533-09/](http://people.cs.ubc.ca/~tmm/courses/533-09/)

- Questions?